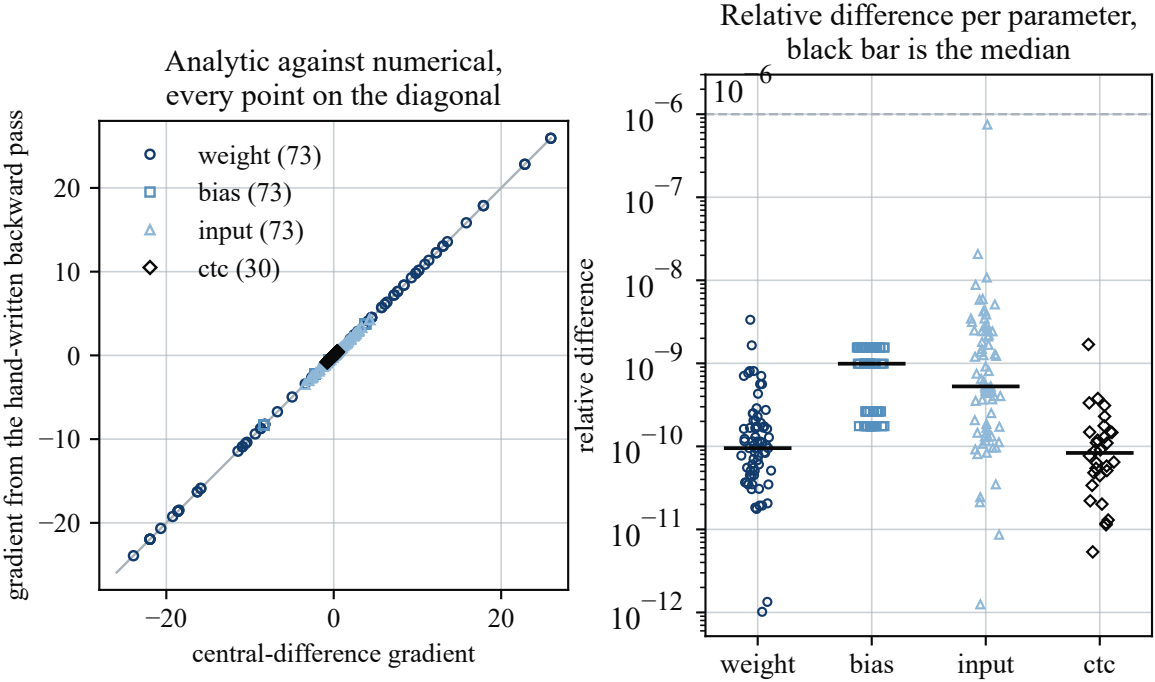
Analytic against numerical,
every point on the diagonal
weight (73)
bias (73)
input (73)
ctc (30)
gradient from the hand-written backward pass
central-difference gradient
20
10
0
−10
−20
−20
0
20
Relative difference per parameter,
black bar is the median
relative difference
10^−6
10^−7
10^−8
10^−9
10^−10
10^−11
10^−12
weight
bias
input
ctc